

Let $\mathbf{x} = [x_0, \dots, x_n]$ be the nodes, and write $p_n(\eta; \mathbf{x})$ when we want to emphasise that the interpolation polynomial depends on the nodes. The cost function is

$$C(\mathbf{x}) := \frac{b-a}{N} \sum_{k=0}^N (f(\eta_k) - p_n(\eta_k))^2, \quad p_n(\eta) = \sum_{i=0}^n \ell_i(\eta) f(x_i),$$

so for $0 \leq r \leq n$

$$\frac{\partial C}{\partial x_r}(\mathbf{x}) = -\frac{2(b-a)}{N} \sum_{k=0}^N (f(\eta_k) - p_n(\eta_k)) \frac{\partial p_n}{\partial x_r}(\eta_k).$$

For the nodes x_i it follows that $p_n(x_i) = f(x_i)$.

Given $i \neq r$. x_r only enters $\ell_i(\eta) = \prod_{m \neq i} \frac{\eta - x_m}{x_i - x_m}$ through the factor with $m = r$, and

$$\begin{aligned} \frac{\partial}{\partial x_r} \left(\frac{\eta - x_r}{x_i - x_r} \right) &= \frac{(-1)(x_i - x_r) - (-1)(\eta - x_r)}{(x_i - x_r)^2} = \frac{\eta - x_i}{(x_i - x_r)^2}, \\ \implies \frac{\partial \ell_i}{\partial x_r}(\eta) &= \ell_i(\eta) \left(\frac{1}{x_i - x_r} - \frac{1}{\eta - x_r} \right). \end{aligned}$$

Given $i = r$. Let

$$\ell_r(\eta) = \prod_{m \neq r} \frac{\eta - x_m}{x_r - x_m} = \frac{\prod_{m \neq r} (\eta - x_m)}{\prod_{m \neq r} (x_r - x_m)} =: \frac{g(\eta)}{h(x_r)}.$$

By the quotient rule,

$$\frac{\partial \ell_r}{\partial x_r} = \frac{0 \cdot h(x_r) - g(\eta) h'(x_r)}{h(x_r)^2} = -\frac{g h'}{h^2}.$$

Since

$$\frac{h'}{h} = \frac{\partial}{\partial x_r} \log |h| = \frac{\partial}{\partial x_r} \sum_{m \neq r} \log |x_r - x_m| = \sum_{m \neq r} \frac{1}{x_r - x_m},$$

we get

$$\frac{\partial \ell_r}{\partial x_r}(\eta) = -\ell_r(\eta) \sum_{m \neq r} \frac{1}{x_r - x_m}.$$

Applying the chain rule in two variables to the identity $p_n(x_r; \mathbf{x}) = f(x_r)$ gives

$$\frac{d}{dx_r} p_n(x_r; \mathbf{x}) = \frac{\partial p_n}{\partial \eta}(x_r; \mathbf{x}) \cdot \frac{\partial x_r}{\partial x_r} + \frac{\partial p_n}{\partial x_r}(x_r; \mathbf{x}),$$

so, writing $p'_n = \partial p_n / \partial \eta$,

$$f'(x_r) = p'_n(x_r) + \frac{\partial p_n}{\partial x_r}(x_r) \iff \frac{\partial p_n}{\partial x_r}(x_r) = f'(x_r) - p'_n(x_r).$$

Now $\frac{\partial p_n}{\partial x_r}$ is a polynomial in η of degree $\leq n$, and $\frac{\partial p_n}{\partial x_r}(x_i) = 0$ for $i \neq r$ (since $p_n(x_i; \mathbf{x}) = f(x_i)$ does not depend on x_r). Hence we can write

$$\frac{\partial p_n}{\partial x_r}(\eta) = (f'(x_r) - p'_n(x_r)) \ell_r(\eta),$$

$$\implies \frac{\partial C}{\partial x_r} = -\frac{2(b-a)}{N} (f'(x_r) - p'_n(x_r)) \sum_{k=0}^N (f(\eta_k) - p_n(\eta_k)) \ell_r(\eta_k).$$

We can now construct ∇C , and find that

$$\nabla C = 0 \iff (f'(x_r) - p'_n(x_r)) \sum_{k=0}^N (f(\eta_k) - p_n(\eta_k)) \ell_r(\eta_k) = 0, \quad 0 \leq r \leq n,$$

which requires, for each r ,

$$f'(x_r) = p'_n(x_r) \quad \vee \quad \langle f - p_n, \ell_r \rangle_N = 0,$$

where

$$\langle g, h \rangle_N := \frac{b-a}{N} \sum_{k=0}^N g(\eta_k) h(\eta_k), \quad \|g\|_N^2 := \langle g, g \rangle_N.$$

The functions ℓ_r , $r = 0, \dots, n$, form a basis for $\mathbb{P}_n$, so if

$$\langle f - p_n, \ell_r \rangle_N = \frac{b-a}{N} \sum_{k=0}^N (f(\eta_k) - p_n(\eta_k)) \ell_r(\eta_k) = 0$$

for all r , then $f - p_n$ is orthogonal to all of $\mathbb{P}_n$.

Let $p \in \mathbb{P}_n$ with $f - p \perp \mathbb{P}_n$. Suppose $f - p$ changes sign on the grid only $m \leq n$ times, namely in the intervals $(\eta_{k_j}, \eta_{k_j+1})$, $j = 1, \dots, m$. Let

$$q(\eta) = \alpha \prod_{j=1}^m (\eta - \beta_j), \quad \alpha = \pm 1, \quad \beta_j \in (\eta_{k_j}, \eta_{k_j+1}).$$

Then $q \in \mathbb{P}_n$ changes sign exactly where $f - p$ does, so with the right choice of α we have $(f - p)(\eta_k) q(\eta_k) \geq 0$ for all k , with strict inequality for at least one k . This is a contradiction, since

$$0 = \langle f - p, q \rangle_N = \frac{b-a}{N} \sum_{k=0}^N (f - p)(\eta_k) q(\eta_k) > 0.$$

Hence $f - p$ changes sign in at least $n + 1$ intervals. Since $f - p$ is continuous and $(f - p)(\eta_{k_j}) \cdot (f - p)(\eta_{k_j+1}) < 0$ on $n + 1$ disjoint intervals $(\eta_{k_j}, \eta_{k_j+1})$, $f - p$ has $n + 1$ distinct zeros in (a, b) .

Consequently $p_n = p^*$, where

$$\|f - p^*\|_N^2 = \min_{q_n \in \mathbb{P}_n} \|f - q_n\|_N^2,$$

i.e. p^* is the unique best approximation polynomial in the (discrete) L^2 .

Requirements for the optimal nodes

$\mathbf{x}$ is optimal if and only if the x_r are distinct zeros of $f - p^*$:

- (I) If all x_r are zeros of $f - p^*$, then p^* interpolates f at the nodes. Uniqueness of the interpolation polynomial gives $p_n = p^*$, so C is minimal.
- (II) Assume $\mathbf{x}$ is optimal. By (I) the minimal value of C is $\|f - p^*\|_N^2$, and p^* is unique, so $p_n = p^*$. Hence $(f - p^*)(x_r) = 0$ for all r .

An alternative method for finding the optimal nodes

Choose an orthogonal basis $\{\varphi_0, \dots, \varphi_n\}$ for $\mathbb{P}_n$ and write $p = \sum_j c_j \varphi_j$. Then

$$\min_{q \in \mathbb{P}_n} \|f - q\|_N^2 = \min_{\mathbf{c} \in \mathbb{R}^{n+1}} \|M\mathbf{c} - \mathbf{y}\|_2^2, \quad M_{kj} = \varphi_j(\eta_k), \quad y_k = f(\eta_k).$$

The least squares method then gives

$$M^T M \mathbf{c} = M^T \mathbf{y} \iff \langle f - p, \varphi_j \rangle_N = 0, \quad j = 0, \dots, n.$$